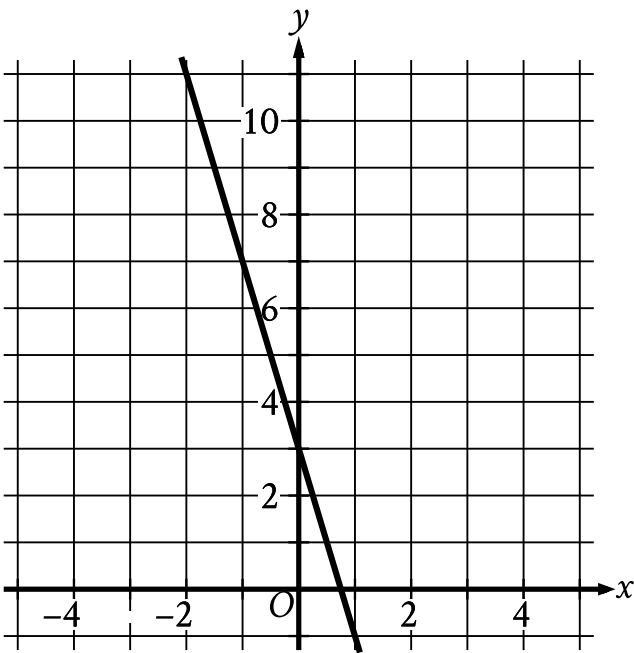


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question



The graph of the linear function $y = f(x) + 19$ is shown. If c and d are positive constants, which equation could define f ?

- Answer
- A. $f(x) = -d - cx$
 - B. $f(x) = d - cx$
 - C. $f(x) = -d + cx$
 - D. $f(x) = d + cx$

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph of the linear function $y = f(x) + 19$ is shown. This means that the graph of $y = f(x) + 19$ can be translated down 19 units to create the graph of $y = f(x)$ and the y-coordinate of every point on the graph of $y = f(x) + 19$ can be decreased by 19 to find the resulting point on the graph of $y = f(x)$. The y-intercept of the graph of $y = f(x) + 19$ is (0, 3). Translating the graph of $y = f(x) + 19$ down 19 units results in a y-intercept of the graph of $y = f(x)$ at the point (0, 3 - 19), or (0, -16). The graph of $y = f(x) + 19$ slants down from left to right, so the slope of the graph is negative. The translation of a linear graph changes its position, but does not change its slope. It follows that the slope of the graph of $y = f(x)$ is also negative. The equation of a linear function f can be written in the form $f(x) = b + mx$, where b is the y-coordinate of the y-intercept and m is the slope of the graph of $y = f(x)$. It's given that c and d are positive constants. Since the y-coordinate of the y-intercept and the slope of the graph of $y = f(x)$ are both negative, it follows that $f(x) = -d - cx$ could define f .

Choice B is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept.

Choice C is incorrect. This could define a linear function where its graph has a positive, not negative, slope.

Choice D is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept and a positive, not negative, slope.